

Predictive Churn Analysis — Logistic Regression

Objective: construct and evaluate a machine-learning classification pipeline for customer retention risk using the uploaded customer churn dataset.

1. Dataset & Preparation

The uploaded dataset contains 15 customers and 12 original columns. The target `Churn` was encoded as 0/1. `CustomerID` was excluded as an identifier. Numeric variables were median-imputed and MinMax-scaled; categorical variables were one-hot encoded with unknown-category handling.

2. Train/Test Split

An 80/20 stratified split with `random_state=42` was used: 12 training rows and 3 test rows.

3. Model Evaluation

Metric	Test-set result
Precision	1.000
Recall	1.000
F1-Score	1.000
ROC-AUC	1.000

4. Churn Risk Scores

The model produces a probability of churn for every customer. Risk bands are defined as Low (<0.33), Medium ($0.33\text{--}0.66$), and High (>0.66) for operational triage. These thresholds are illustrative and should be tuned using business costs and validation data.

Top 5 predicted-risk customers

CustomerID	Churn Probability	Risk Band	Actual Churn
CUST-1003	86.4%	High	Yes
CUST-1007	86.0%	High	Yes
CUST-1014	83.5%	High	Yes
CUST-1005	82.9%	High	Yes
CUST-1010	81.8%	High	Yes

5. Reproducibility

The accompanying Jupyter Notebook contains the full preprocessing, 80/20 split, Logistic Regression model, evaluation metrics, ROC curve, and exported customer risk scores.

6. Limitation

The dataset contains only 15 customers, so the 20% test set has only 3 observations. Holdout Precision, Recall, F1 and ROC-AUC are therefore highly unstable and should not be treated as production-quality performance estimates. A larger historical dataset and cross-validation should be used before deployment.